WORKFORCE INTELLIGENCE ANALYTICS PLATFORM

End-to-End Technical Documentation & Insight Report

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Introduction

This project presents the design and delivery of a Workforce Intelligence Analytics Platform developed to provide a unified view of workforce performance across hiring, retention, engagement, learning, and operations.

The objective is not simply to build dashboards, but to create a complete analytical solution that connects data engineering, modelling, and business insight into a single, structured platform. The solution brings together multiple HR data domains including HRIS, Applicant Tracking Systems, Learning Management Systems, engagement surveys, and timekeeping data. These sources are integrated, transformed, and modelled to enable consistent and reliable reporting.

The final output (Business intelligence dashboard and insights) supports key stakeholders including the CHRO, VP People, Talent Acquisition Leads, and HR Business Partners by providing clear visibility into workforce risks and performance drivers.

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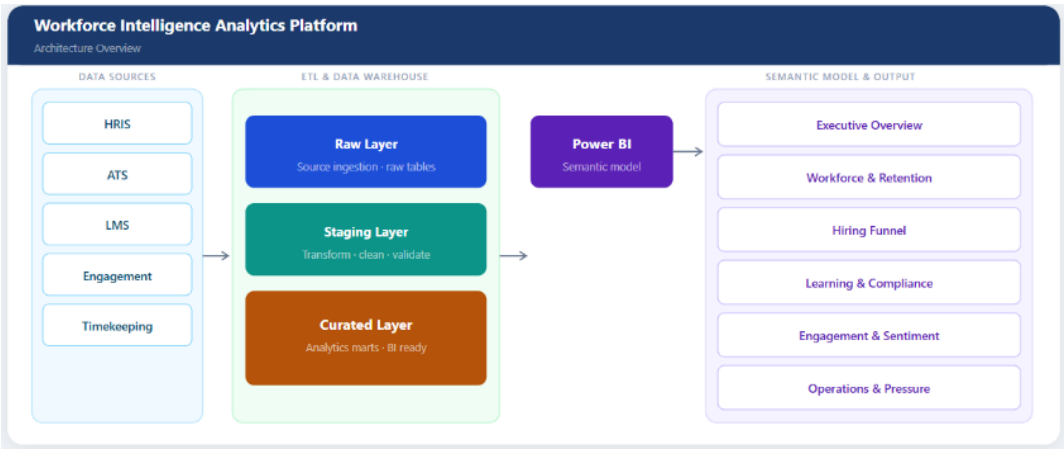
Platform Architecture and Design



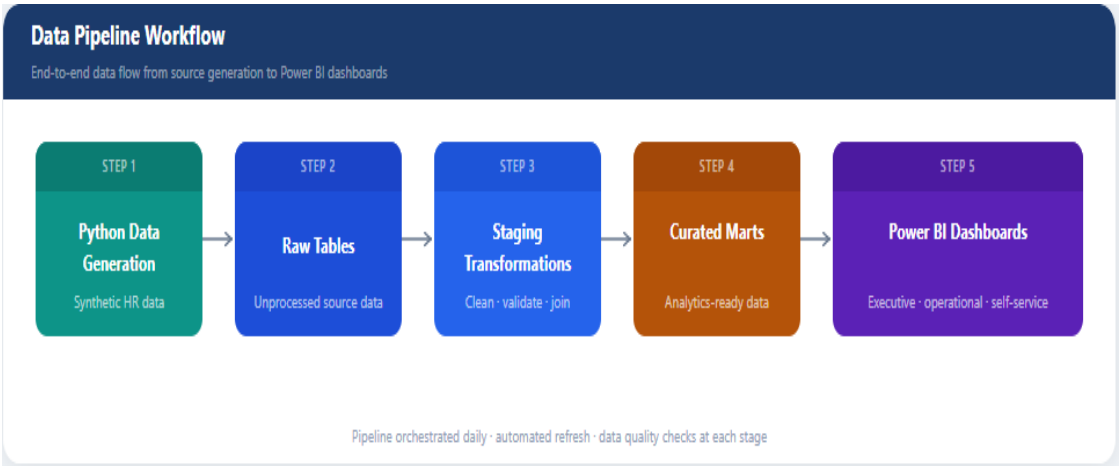
The platform follows a layered data architecture designed to ensure scalability, traceability, and clarity across the entire data lifecycle. Data is ingested from multiple source systems into a raw layer, where it is stored without transformation. This preserves data integrity and allows for reproducibility.

A staging layer is then applied to clean, standardise, and validate the data. This includes handling missing values, aligning schemas across systems, and preparing datasets for integration.

The curated layer contains analytics-ready data marts. Each mart is designed around a specific business domain such as hiring, engagement, learning, operations, and workforce summary. These marts serve as the foundation for reporting and analysis.

This structured approach ensures that all downstream insights are built on consistent and validated data.

Data Pipeline and Transformation

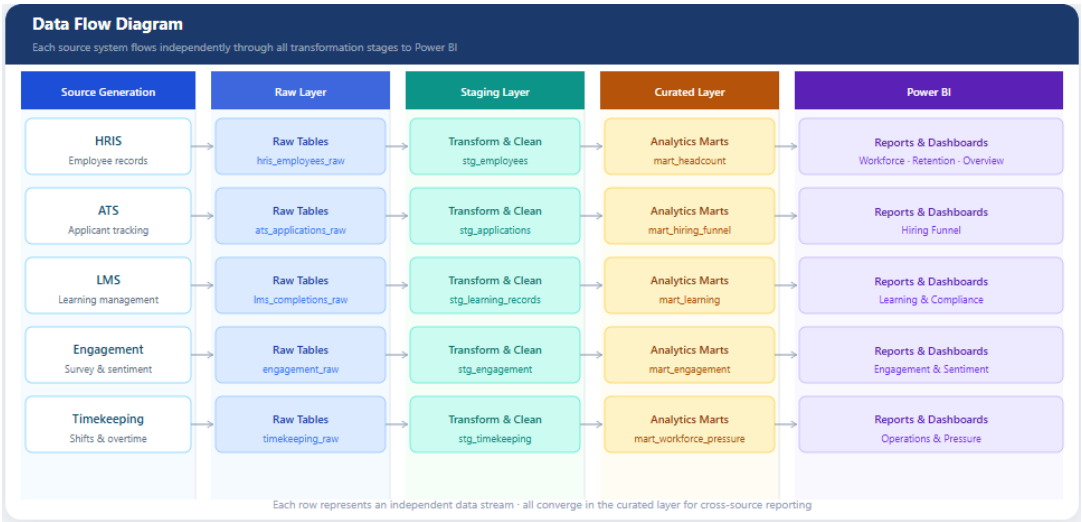


The data pipeline is designed as an end-to-end workflow that moves data from generation to final reporting. Synthetic HR data is generated using Python to simulate realistic workforce scenarios. This includes hiring pipelines, employee records, engagement scores, and operational metrics.

SQL transformations are used to move data through each stage of the pipeline. Raw tables are loaded, cleaned in staging, and transformed into curated marts using well-defined business logic.

Data quality checks are applied at each stage to ensure accuracy and consistency. These checks validate record counts, null handling, and metric calculations. This pipeline structure ensures that the data used in reporting is both reliable and explainable.

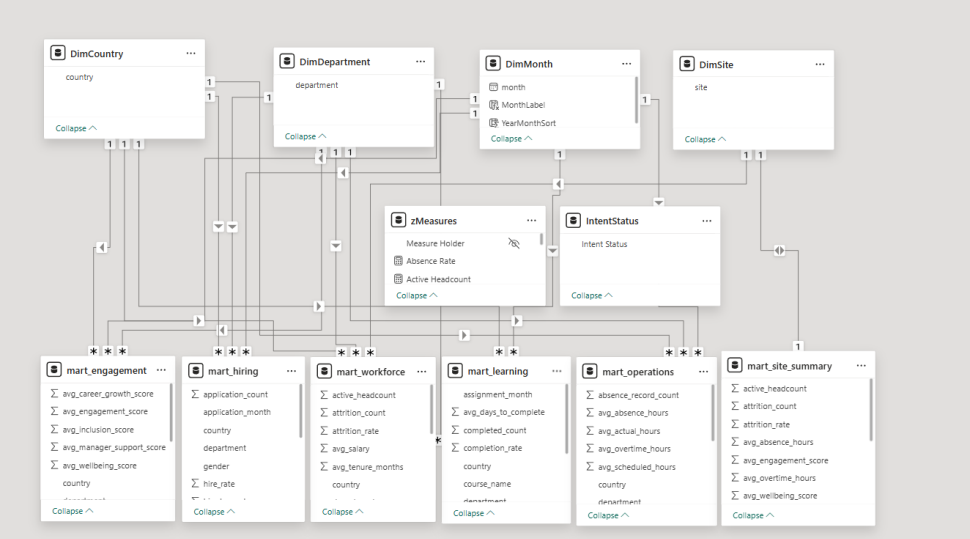
Data Flow and Integration



Each data source flows independently through the transformation layers before being unified in the curated layer. This design allows for flexibility and modularity. Changes in one data source do not disrupt the entire pipeline, and each domain can be maintained independently.

At the curated layer, datasets are aligned across shared dimensions such as department, country, and time. This enables cross-functional analysis across hiring, engagement, learning, and operations.

Data Modelling Approach



A star schema model was implemented in Power BI to support efficient reporting and filtering. Fact tables include hiring, engagement, learning, operations, and workforce metrics. These are connected to shared dimension tables such as country, department, site, and time.

A dedicated measures table was created to centralise all calculations. This ensures consistency in metric definitions and improves model usability. The model is designed to support both high-level executive reporting and detailed operational analysis.

Workforce Insights and Analysis

The analysis focuses on understanding workforce performance across key dimensions and identifying areas of risk.

Workforce Stability and Attrition

Attrition is recorded at 18.5 percent, which is elevated and indicates a potential retention issue. This is not evenly distributed across the organisation. Operational and support functions show higher attrition levels compared to corporate functions. This suggests that certain roles may be experiencing higher levels of pressure, lower engagement, or limited career progression.

Engagement and Flight Risk

Average engagement levels are approximately 60, with wellbeing slightly lower. While this does not indicate a disengaged workforce, it highlights a level of vulnerability.

Analysis shows a clear relationship between declining engagement and increased intent to leave. Departments with lower engagement scores consistently show higher flight risk. This confirms that engagement is a leading indicator of attrition.

Hiring Performance

The organisation demonstrates strong hiring activity, with 1080 hires recorded. However, conversion from application to hire is approximately 13.5 percent. This indicates inefficiencies in the recruitment process. A high volume of applications is not translating into proportional hiring outcomes.

Potential issues may include screening bottlenecks, interview conversion challenges, or misalignment between candidate profiles and role requirements.

Learning and Compliance

Learning completion stands at 64.2 percent, with a significant volume of overdue training. This represents both a compliance risk and a capability gap. Departments with lower completion rates may lack the necessary training to perform effectively, which can contribute to performance issues and attrition.

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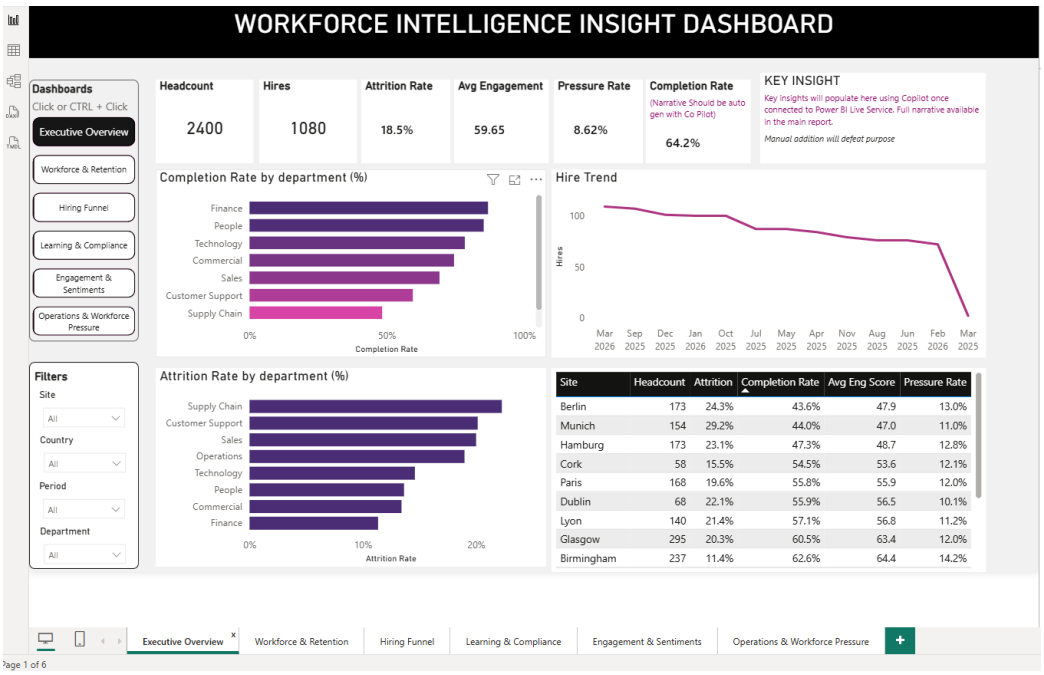
Automated Insight Capability

Power BI provides a narrative feature that can automatically generate text-based insights based on the current state of a report page. This includes dynamic updates when filters or slicers are applied, allowing the narrative to reflect changes in metrics in real time. This functionality was considered during development as a way to enhance automated insight delivery. However, it is only available within higher-tier Power BI environments and was not accessible within the scope of this project.

As a result, insight generation remains supported through the dashboard visuals, while detailed interpretation is provided within this document.

Dashboard Outputs



The Power BI dashboards provide a structured view of workforce performance across six key areas. Each page is designed to answer a specific business question, from overall workforce health to detailed drivers of attrition and operational strain. Interactive filtering allows stakeholders to explore data by department, country, and time period, enabling both high-level and detailed analysis.

Key Business Conclusions

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Final Remarks

The organisation demonstrates strong capability in attracting talent but faces challenges in retaining and supporting its workforce.

Hiring performance is not matched by retention outcomes, leading to a cycle where new hires replace departing employees rather than contributing to net growth.

Engagement and wellbeing levels suggest that employees are not fully disengaged but are at risk, particularly in high-pressure environments.

Learning compliance gaps further contribute to this risk by limiting workforce readiness and development.

Operational pressure is a critical factor, particularly in specific departments where workload and absence levels are higher.